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EFFECT OF SURFACE TEMPERATURE ON THE SORPTION OF HYDROGEN BY Pd(111) *

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Temperature programmed desorption (TPD) subsequent to various hydrogen exposure conditions indicates the formation of chemisorption, solid solution, and hydride phases of hydrogen in the near surface region of Pd(111). Variation of the sample exposure temperature (T_e) between 80 and 300 K has a strong effect on the subsequent TPD spectra. At $T_e = 80$ K a single desorption peak, β, appears at 310 K. Coverage variation of the β peak is consistent with second-order recombinative desorption of chemisorbed hydrogen. For T_e between 90 and 140 K a slight enhancement of the β peak occurs and a new peak, α, appears initially near 170 K. It does not saturate, exhibits near-zeroth-order desorption kinetics, and is assigned to the decomposition of a near surface palladium hydride phase. Population of the α peak is thermally activated with a maximum at $T_e \approx 115$ K. For T_e greater than 140 K, α disappears while the total amount of absorbed hydrogen increases significantly. At these temperatures, the concentration of absorbed hydrogen decreases significantly if the sample is held in vacuo at T_e after completion of the hydrogen exposure. At all exposure temperatures there is also a broad desorption feature near 800 K which is enhanced by higher T_e and is associated with hydrogen in solid solution.

There have been numerous surface science studies on the interaction of hydrogen with metals. Many of these have been on the group VIII and IB transition metals which play an important role in catalytic reactions and in the technology of modern high strength metallurgical alloys. Of the twelve metals in these groups, only palladium reacts with hydrogen exothermically to form a stable hydride phase [1]. It is therefore not surprising that the interaction of hydrogen with Pd surfaces is different, when compared, to neighboring transition metals. In particular, evidence for near-surface penetration has been reported in molecular beam [2], thermal desorption [3,4], He-scattering [5], photoemission [6,7], low energy electron diffraction [8,9], and electron-stimulated desorption [10] studies. However, a comparison of previous results by various authors is complicated by the fact that in each case, the H/Pd experiments were carried out over a different range of temperature and exposure conditions.

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Previous temperature programmed desorption (TPD) studies of the interaction of hydrogen with different palladium surface orientations have been carried out for sample temperatures between 100 and 300 K. Initial exposure to hydrogen leads to desorption between 300 and 400 K. This desorption peak, designated β, has been associated with the recombination of chemisorbed hydrogen. At higher exposures (~ 5 langmuirs; 1 L = 10^{-6} Torr s) and sample temperatures of 300 K, absorption of hydrogen has been reported in Pd(111) [11,12], Pd(110) [11], and polycrystalline palladium ribbon [13]. In TPD this absorbed hydrogen desorbs at higher temperatures than the β peak. For exposures below room temperature, hydrogen penetration into the bulk has also been reported for polycrystalline palladium wire at 100 K [14,15], for Pd(100) at 170 K [16], and for Pd(110) at 120 K [3–5]. The TPD measurements for the palladium wire as well as Pd(110) showed that a desorption peak, designated α, occurred at lower temperatures than the β peak. The α peak was associated with near-surface hydrogen and for Pd(110) [3–5] was ascribed to hydrogen at sites made accessible by hydrogen induced surface reconstruction.

In the present study, TPD was systematically used to monitor the effect of exposure conditions, in particular, sample temperature, on the sorption (both adsorption and absorption) of hydrogen by Pd(111). It is shown that hydrogen penetrates into the near surface region of Pd(111) at temperatures as low as 90 K without any surface reconstruction. It is also shown that the particular phase of hydrogen formed in the near surface region is strongly dependent on the exposure parameters and that at certain exposure temperatures the hydrogen phase formed is unstable upon removal of gas phase H_2. Finally, we suggest that the general features observed here should manifest themselves on all palladium surfaces.

The experiments were performed in a stainless steel ultrahigh vacuum chamber with a base pressure of 8×10^{-11} Torr. The 8 mm diameter Pd(111) sample was first cleaned by repeated argon ion sputtering and high temperature annealing cycles. Additional exposure to oxygen at sample temperatures above the CO desorption temperature (~ 500 K) but below that for oxygen desorption (~ 800 K) was used to clean off residual carbon. The surface was judged free of carbon when no CO desorption peak was observed in TPD after low temperature oxygen exposures. Surface cleanliness was also checked with Auger electron spectroscopy (AES). The TPD spectra were obtained using a differentially pumped mass spectrometer with a 5 mm entrance aperture positioned to within less than 0.15 mm of the sample surface. Sample temperature was monitored by a chromel–alumel thermocouple spot-welded to the front surface of the Pd(111) crystal. The low temperature response of this thermocouple was determined by calibrating to a known iron-doped gold versus chromel thermocouple standard. The sample was mounted onto an indirect heater assembly, which in turn was attached to a copper cold finger.

This assembly was capable of cooling from 300 K to less than 85 K in 100 s and could be heated at rates between 4 and 10 K/s. For all TPD experiments exposures were obtained by back-filling the chamber; a measured correction factor of 2.3 was used to determine the actual hydrogen pressure from the ion gauge reading. The sample was surrounded by a liquid nitrogen cooled copper cylinder during exposure to eliminate coadsorption of other gases: no CO or H_2O was found to desorb during TPD. After exposure the sample was immediately quenched unless otherwise stated and was annealed to 1000 K between each run to ensure that all of the hydrogen was removed from the bulk [13].

The H_2 TPD spectra obtained subsequent to H_2 exposure at 80 K are shown in fig. 1. A single desorption peak, β, whose temperature shifts from 370 K at low coverage to 310 K at high coverage was observed. This desorption peak results from recombination of chemisorbed atomic hydrogen and is consistent with previous work which indicated that the adsorption process was dissociative [11,12,17]. An intriguing aspect of this chemisorbed state for the Pd(111) surface is that it involves occupation of octahedral sites between the first and second layers as well as surface threefold hollow sites [8,9].

Whereas the present observation of a single desorption peak for chemisorbed hydrogen is in agreement with recent work of Kok et al. [18,19], it is in

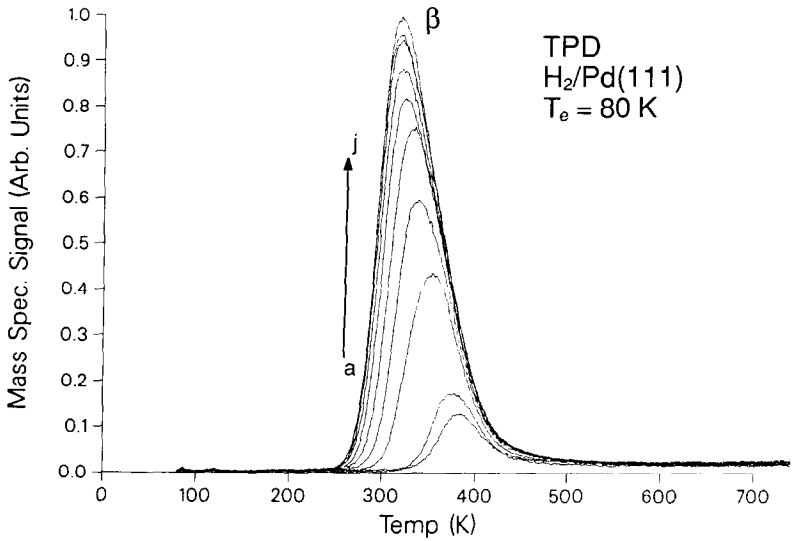


Fig. 1. TPD spectra for H_2/Pd(111) subsequent to hydrogen dosing at 80 K. The exposures were obtained by back-filling the chamber. The heating rate was 5.8 K/s. Exposures are: (a) background, (b) 0.07, (c) 0.34, (d) 0.69, (e) 1.04, (f) 1.38, (g) 1.72, (h) 2.07, (i) 2.07, and (j) 3.45 L. Exposures (h) and (i) demonstrate level of reproducibility.

contrast to the results of Kiskinova and Bliznakov [12] who recently reported low temperature hydrogen TPD in which H_2 desorbed in a two peak structure for adsorption at 250 K. In both studies the authors discounted the possibility of bulk absorption at sample temperatures below 300 K. As shown below, hydrogen bulk penetration can occur at temperatures as low as 90 K and in fact, quite readily above 200 K.

Large exposures (> 20 L) with the sample exposure temperatures (T_e) between 90 and 140 K result in the emergence of a new TPD peak, α, at 150 K. The rate of population of α initially increases with increasing T_e, reaching a maximum near 115 K. Roughening the surface by Ar ion sputtering enhances the rate of population of the α peak by up to a factor of 4 at 115 K. Fig. 2 is a series of desorption spectra following various exposures at $T_e = 115$ K. With increasing exposure, the α peak grows, shifts to higher temperatures, and, above 240 L, approaches zeroth-order desorption kinetics (i.e. the leading edges of the TPD spectra begin to overlap). The α peak does not appear to saturate at an exposure of 10000 L, which was the largest used. At this exposure the α peak area is about twice as large as that of β as can be seen from fig. 2. Also apparent in fig. 2 is that desorption in β is slightly enhanced as α develops. The LEED pattern obtained after exposures sufficient to produce the α peak exhibit (1×1) symmetry, indicating no concomitant surface reconstruction. A $(\sqrt{3} \times \sqrt{3})$R30° LEED structure is observed at hydrogen coverages less than 2/3 monolayer and temperatures less than 110

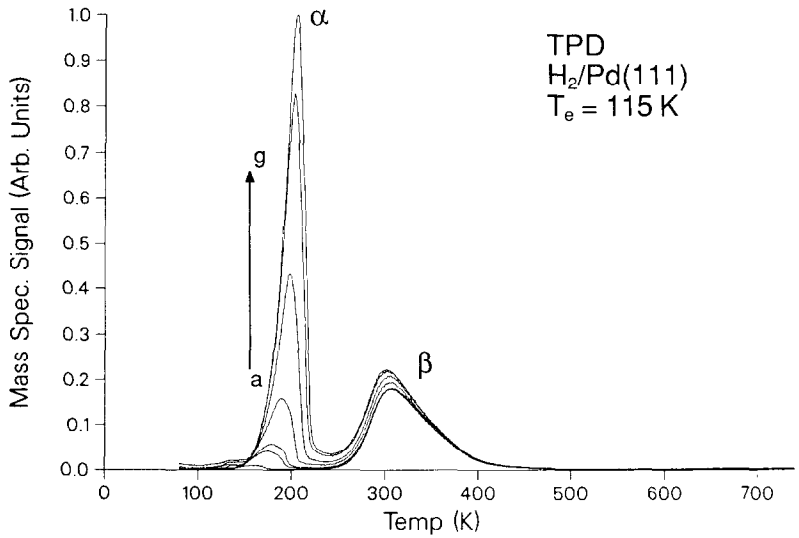


Fig. 2. TPD spectra for H_2/Pd(111) subsequent to hydrogen dosing at 115 K. The exposure pressure was 2.8×10^{-6} Torr and the heating rate was 3.5 K/s. Exposures are: (a) 166, (b) 828, (c) 1159, (d) 2484, (e) 4968, (f) 7452, and (g) 9936 L.

K, but it is due to the chemisorbed hydrogen and not a surface reconstruction [8,9,20].

The α peak in the present study behaves similarly to the α peaks observed on other Pd surfaces [3,13,14]: it appears only after large exposure, desorbs at a temperature below that for chemisorbed hydrogen and exhibits near-zeroth-order desorption kinetics. These similarities suggest a common origin for this peak, namely absorbed hydrogen in the near surface region. In addition, the near-zeroth-order desorption kinetics requires a hydrogen reservoir to supply hydrogen rapidly to the surface, where recombination and desorption can occur. The desorption characteristics of the α peak require that the absorbed phase is highly concentrated near the surface. In bulk palladium at temperatures < 300 K there are two phases of absorbed hydrogen: solid solution which occurs at low concentrations, $H/Pd \leq 0.01$, and hydride at high concentrations, $H/Pd \geq 0.6$. For $0.01 < H/Pd < 0.6$, regions of the hydride phase coexist with the solid solution phase [1]. The α peak is then assigned to the decomposition of near surface hydride. The hydride forms at low T_e because hydrogen diffusion away from the surface is extremely slow and concentration of hydrogen near the surface occurs. The slow hydrogen diffusion away from the surface may also be due in part to near surface lattice strain effects which may stabilize the hydride phase [1]. Since the hydride is localized near the surface, hydrogen transport to the surface during the hydride decomposition should be rapid.

The effect of exposure temperature on the TPD spectra is shown in fig. 3. The spectra were obtained by holding the sample at a constant temperature, T_e, during the exposure, quenching the sample to 80 K while simultaneously pumping out the hydrogen gas, and then heating for the TPD experiment. The hydrogen exposures were fixed at 630 L (H_2 exposure pressure: 2.8×10^{-6} Torr) and T_e ranging from 80 to 298 K. Even at these large exposures, for $T_e = 80$ K only the β peak appears in the TPD spectrum. As stated above, raising T_e above 90 K results in the appearance of the α peak, maximizing near 115 K (see curve b). Above 115 K, population of α decreases but the total uptake of hydrogen by the sample still increases. At $T_e = 150$ K the α peak disappears, and a small low temperature shoulder appears on the β peak which in turn becomes larger and shifts to slightly lower temperature (see curve c). Raising T_e to 190 K eliminates the low temperature shoulder on the β peak. At $T_e = 224$ K (curve d) the β peak broadens and further shifts to slightly lower temperature. Above $T_e = 224$ K the area of the β peak continues to increase primarily on the high temperature side. A maximum in the β peak height occurs at $T_e = 261$ K (see curve e). Above this T_e, the β peak shifts to higher temperatures and its height decreases, although the area under the entire TPD spectrum is still increasing. Desorption on the high temperature side of β increases significantly as T_e is further increased to 298 K (curve f). For all the spectra there is an additional broad desorption feature at 800 K;

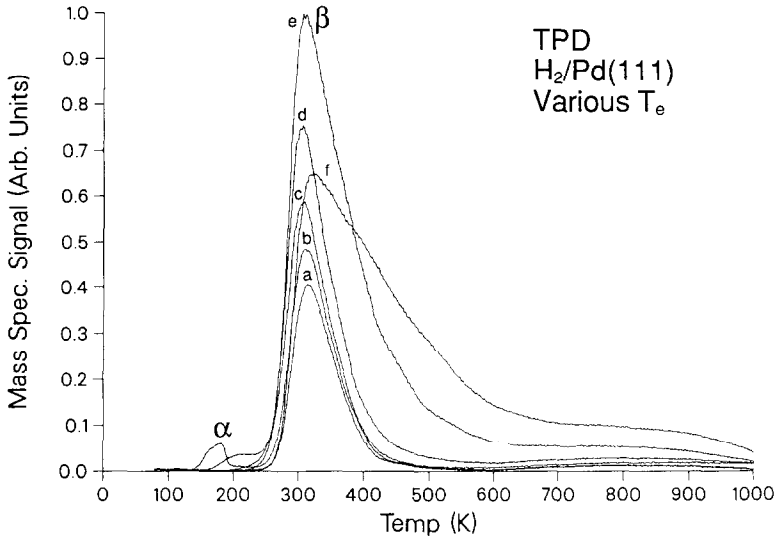


Fig. 3. TPD spectra for H_2/Pd(111) subsequent to a 630 L exposure of hydrogen at various sample temperatures. The exposure pressure was 2.8×10^{-6} Torr and the heating rate was 5.9 K/s. The sample temperatures were: (a) 80, (b) 115, (c) 150, (d) 224, (e) 261, and (f) 298 K.

population of this feature increases with increasing T_e. As previously mentioned, sputtering the Pd(111) surface causes an increase in hydrogen absorption rate at $T_e = 115$ K. This enhancement in absorption rate for the sputtered surface decreases as T_e is increased. In fact, at $T_e = 298$ K there is little difference in the hydrogen uptake between a high temperature annealed surface and a sputtered surface.

In order to determine whether the desorption spectra in fig. 3 represent stable Pd hydrogen phases, desorption spectra following the same exposure were obtained after annealing the sample at the exposure temperature for 600 s subsequent to removal of the gas phase H_2. For T_e below 200 K little or no change in the desorption spectra occurs. Above $T_e = 200$ K there is a large decrease in the hydrogen desorbing in the region between the β peak and 700 K, through there is little if any change in the region of the broad desorption feature at 800 K. An example of this effect is illustrated in fig. 4 which shows the desorption spectra for $T_e = 298$ K with and without annealing. Without annealing, the desorption spectrum is representative of the hydrogen uptake obtained during the exposure at 2.8×10^{-6} Torr which is "frozen in" by quenching. With annealing, the dissolved hydrogen tends to equilibrate with the much lower residual hydrogen partial pressure after pump out ($(1-2) \times 10^{-10}$ Torr) resulting in a loss of hydrogen from the metal. Based on an activation energy for bulk diffusion equal to 0.23 eV and a preexponential factor equal to 2.9×10^{-3} cm^2/s [1], hydrogen diffusivity at 298 K equals

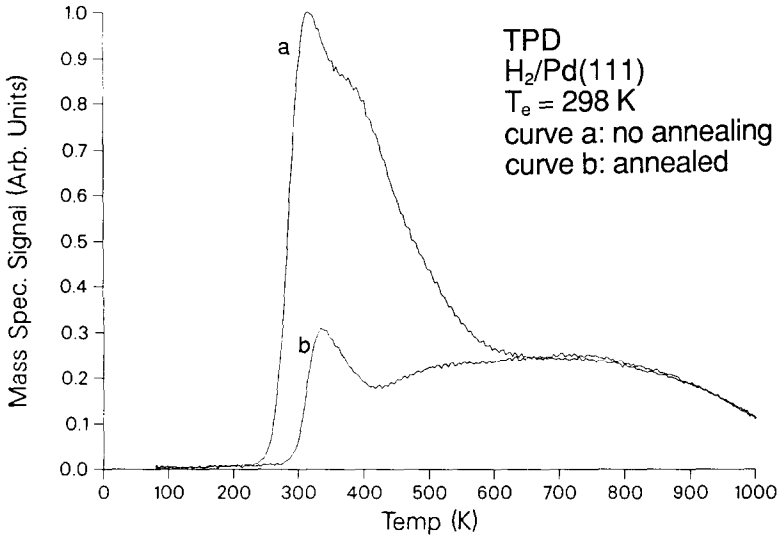


Fig. 4. TPD spectra for H_2/Pd(111) following 630 L doses at 298 K. The heating rate was 5.9 K/s. Curve (a): the sample was quenched simultaneously with removal of gas phase H_2. Curve (b): the sample was annealed at 298 K for 600 s following removal of gas phase H_2. The annealing released almost 1/2 of the original uptake of hydrogen.

3.8×10^{-7} cm^2/s and the annealing procedure associated with the data in fig. 4 is sufficient to allow release of hydrogen absorbed as deep as 0.015 cm below the surface. It is this hydrogen that accounts for the difference in the two TPD spectra in fig. 4. The broad high temperature feature centered near 800 K which is still apparent after the anneal must come from hydrogen in solution deep within the bulk as previously proposed by Conrad et al. [11]. The formation of the unstable hydrogen phase implies a pressure dependence of hydrogen uptake by the sample which will be fully reported elsewhere [21].

Fig. 5 shows the hydrogen uptake by the Pd(111) sample as a function of T_e. The uptake is given in terms of monolayers, where a monolayer is defined as saturation of the β peak at 80 K. The data were taken from the spectra shown in fig. 3 which were obtained by rapid quenching in order to freeze in the distribution at the exposure conditions. The amount of hydrogen uptake increases with increasing T_e, consistent with a kinetically limited absorption process. If, in contrast, equilibrium were to prevail at the exposure conditions, then the hydrogen uptake would decrease with increasing T_e since hydrogen absorption is exothermic in Pd [1]. The small initial increase in uptake between 80 and 140 K is primarily due to increasing population of the α peak. Beyond $T_e = 200$ K the curve rises more rapidly indicating that hydrogen penetration into Pd(111) becomes relatively facile for $T_e > 200$ K. A similar conclusion was obtained from electron-stimulated desorption results on H_2/Pd(111) [10].

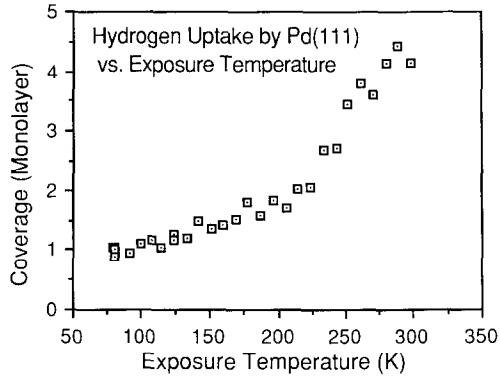


Fig. 5. Hydrogen uptake as a function of exposure temperature. A monolayer is defined as saturation of the β peak at 80 K. The data were obtained from TPD spectra such as those in fig. 3.

In summary, the characteristic TPD spectrum for H/Pd(111) is strongly dependent on (a) the sample temperature, T_e, during exposure and (b) the temperature history of the sample after exposure but prior to TPD. At low temperatures ($T_e < 90$ K), extended hydrogen exposure leads only to a saturation of the chemisorbed surface layer without any significant bulk penetration. Between approximately 90 and 140 K, large hydrogen exposures lead to the formation of a near surface hydride which initially desorbs near 170 K. The non-saturating population indicates that the hydride is not limited to the first sublayer of the sample. For exposures above approximately 140 K and pressures in the 10^{-6} Torr range, hydrogen is absorbed by the sample, but a near surface hydride is *not* formed since the hydrogen can diffuse into the bulk rapidly and a critical concentration for surface hydride formation is not reached. Above $T_e = 200$ K, large exposures lead to significant uptake of hydrogen by the sample which leads to changes in the TPD spectra in the region of the β peak and above ($300 < T_e < 1000$ K). At these warmer exposure conditions, the bulk hydrogen concentration profile is clearly non-equilibrium so that resulting TPD spectra are strongly dependent on the temperature history subsequent to the hydrogen exposure.

Finally, nearly all of the variations in the present TPD spectra are characteristic of hydrogen absorption into bulk palladium. As such these effects might be expected for all palladium samples regardless of surface crystallography or morphology. Indeed the α peak has been previously seen on Pd wire [14,15] and Pd(110) [3–5] and the 800 K peak has been seen for room temperature adsorption on Pd(110) [11]. However, the *rate* at which the Pd(111) surface absorbs hydrogen at low temperatures is extremely slow in comparison to other Pd surfaces [3–5,14,15]. In fact, the results presented here

may represent only an upper limit to the absorption rate for the (111) surface since we cannot rule out the possibility that absorption at low temperatures is controlled primarily by defects.

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